Swara

Swara is an open source project, originally written as a research project by students and professors at MIT to augment the outreach and activities of CGNet, a people's discussion group working with indigenous communities in central India. CGNet was started by veteran journalist Shubhranshu Choudhary, who returned to Central India, where he grew up, to find it torn by violence. Probing to find the reason for the conflict, he quickly realised that open, accessible community media would be a key component of any solution to the conflict. Given that Internet penetration in the region is less than 1 per cent, and community radio is limited by regulation, the next best medium for a community platform was the mobile phone.

The first pilot of Swara was deployed in Bengaluru for use by indigenous communities in Chhattisgarh and neighbouring states in February 2010. Today, the pilot receives over 300 calls a day, and the team is working on building the platform out as an open source project for deployment in other locations. The first replica of the project went live in Indonesia in December 2011.

Swara uses a combination of the Asterisk PBX system in combination with the LoudBlog audio blogging platform, with the integration written in Python. The tested interfaces are GSM gateways (Topex Mobilink, etc) and fixed lines (PRI/BRI) using a Digium telephony card.

FreedomFone

FreedomFone was developed by Alberto Escudero Pascual and Louise Berthilson of IT46, a Swedish IT consultancy, for the Kubatana Trust in Zimbabwe. It was created for many of the same reasons as Swara was developed in India, i.e., lack of impartial and open commercial media, and the need for local and community-level reporting. The FreedomFone pilot, a weekly audio magazine called Inzwa, has been running in Zimbabwe since July 2009, and received over 2500 calls between July and September 2009. FreedomFone's team is also working on developing the platform as a user-friendly DIY IVR kit, and is keen on replicating the model in other areas.

FreedomFone uses the FreeSWITCH soft switch to interface with telephony devices such as the Mobigater and Office Router GSM gateways. The content management system is written in CakePHP, and FreedomFone additionally uses the Cepstral speech synthesis system for text-to-speech conversions. The stated objective is to create a purely phone-accessible platform.

Deployment 101

Both platforms have an almost identical design, as would most audio portal software. This is almost analogous to how traditional websites are built, with the choice of platform being similar to the choice between different Web frameworks. Just as you will find lots of different opinions and preferences for Web platforms among Web designers, you will find that the few implementers of audio portals are just as varied in their preferences for platforms. This usually depends on which

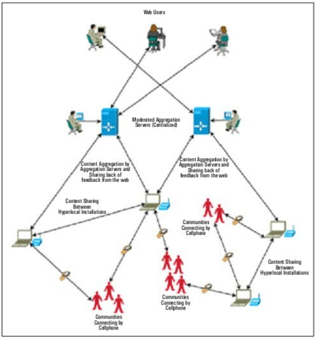

Figure 5: The advantage of hyperlocal deployments

platform the implementer is most familiar with—and if you are implementing your own, one is essentially as good as the other.

The key question, irrespective of which platform you use, is one of deployment strategy. At present, most implementations of audio portals as community media platforms are centralised instances deployed by a single organisation or group, with a specific agenda (such as news, healthcare or governance).

Centralised function-oriented deployment

Centralised, function-oriented deployments require content of a certain quality and, as a result, must usually be moderated. Speech-recognition technology, particularly in the area of automatic transcription, is still a far cry from being very accurate. As a result, moderating a function-specific audio portal is still a manual job, for the most part. Typically, audio portal moderators will need to listen to each message and summarise and/or transcribe it. Beyond transcription, there may be more work to do to improve the quality of the content for the specific purpose of the deployment, like sound quality clean-ups and edits, fact verification (if journalism is the function, for example) and categorisation. All of this work is further exacerbated in a centralised deployment, since all incoming calls come to the same central hub (see Figure 4). In India, and other countries where long-distance call charges are higher than local call charges, centralised platforms also suffer from an added cost element, since all callers must call the central number, regardless of their own locations.

Hyperlocal deployments

An alternative model is a hyperlocal community-oriented one. In this model, an instance of the platform is deployed at the community level and maintained by community members. Such